Higher National Diploma in Engineering (Civil)
Higher National Diploma in Quantity Surveying
First Year, First Semester Examination – 2018
CE1103 - Engineering Surveying I / QS11033 – Fundamentals of Engineering Surveying

Instructions for Candidates:	No. of questions : 06
Answer any **Five (05)** questions.	No. of pages : 03
All question carry equal marks	Time : Three (03) hours

1.

i. Briefly explain the importance of surveying in the construction field (05 marks)

ii. Write short notes on two basic principle of surveying. (05 marks)

iii. What is the purpose of significant figures? (02 marks)

iv. Find the number of significant figures of followings

a. 21.10 b. 0.018 c. 305 (03 marks)

v. Briefly discuss the types of distance measurement with examples (05 marks)

(Total 20 marks)

2.

i. Briefly discuss the followings with the reasons and the methods of eliminating in linear measurements.

a. Instrumental error (02 marks)

b. Natural error (06 marks)

c. Personal error (02 marks)

ii. A steel tape was exactly 30m at the temperature 20°C and under a pull of 100N. A survey line was measured with this tape at a temperature of 30°C and under a pull of 150N and the survey line found to be 780m. The cross sectional area of the tape is 3×10^{-6} m^2, its total weight is 0.690 kg, Young's modulus $E = 2.1 \times 10^4$ N/mm^2, coefficient of thermal expansion $\alpha = 11 \times 10^{-6}$ per °C, g = 9.81 m/s2. If the tape was permanently stretched by 2mm. Find the true length.

(10 marks)

(Total 20 marks)

3..

i. Briefly explain the following terms.

a. Main survey stations (02 marks)

b. Check lines (02 marks)

c. Offsets (02 marks)

d. Chainage (02 marks)

e. Conventional symbols (02 marks)

ii. Following data refers to a plot of a land which is surveyed using a single survey line.

a. Enter these data on a field book page using double line entry method (05 marks)

b. Find the area of the land plot. (05 marks)

Chainage (m)	**Offsets (m)**
0 (Station A)	0.0
5	8.0 Right
12	10.0 Left
25	6.0 Right
30	7.0 Left
45	4.0 Right
50 (Station B)	0.0

(Total 20 marks)

4.

i. Briefly explain the automatic data collector and its advantages and disadvantages (05 marks)

ii. Briefly explain the importance of field book and its types. (06 marks)

iii. Differentiate the main three types of EDM instruments. (09 marks)

(Total 20 marks)

5.

i. Define the words **accuracy** and **precision** (04 marks)

ii. A length was measured 10 times as follows 2.54, 2.53, 2.51, 2.56, 2.57, 2.54, 2.54, 2.55, 2.52, 2.50

Find the followings;

a. Residuals

b. Standard deviation

c. Probable error of a single measurement

d. 90% Error

e. 95% Error (10 marks)

iii. Comment on the measured lengths with the above calculations. (02 marks)

iv. What do you understand by E_{90}, E_{50} error (04 marks)

(Total 20 marks)

6.

i. What are the common instruments that are used in chain surveying? (03 marks)

ii. Briefly explain a method of taking horizontal measurements on a uneven ground (05 marks)

iii. Briefly explain two methods of dropping right angle from a point to a chain (perpendicular offsets) (04 marks)

iv. What are the main considerations at the selection of main survey stations (03 marks)

v. Briefly explain a method of chaining across a narrow river (vision is possible between two banks but the chaining is impossible) (05 marks)

(Total 20 marks)